Introduction to Algebra

Algebra

Algebra is a branch of mathematics that uses letters and symbols to represent numbers.
These letters are called variables.

Variable

A variable is an unknown quantity that is prone to change with the context of a situation.

Example: In the expression 2x+5, x is the variable.

Constant

Constant is a quantity which has a fixed value. In the given example 2x+5, 5 is the constant.

Terms of an Expression

Parts of an expression which are formed separately first and then added or subtracted, are known as terms.

In the above-given example, terms 2x and 5 are added to form the expression (2x+5).

Coefficient of a Term

The numerical factor of a term is called the coefficient of the term.

In the above-given example, 2 is the coefficient of the term 2x.

To know more about Term, factor and Coefficient,

Like and Unlike Terms

Like Terms

Terms having the same variables are called like terms.

Example: 8xy and 3xy are like terms.

Unlike Terms

Terms having different variables are called, unlike terms.

Example: 7xy and -3x are unlike terms.

Monomial, Binomial, Trinomial and Polynomial Terms

Name	Monomial	Binomial	Trinomial	Polynomial
No. of terms	1	2	3	>3
Example	7xy	(4x−3)	(3x+5y−6)	(6x+5yx−3y+4)

Formation of Algebraic Expressions

Combinations of variables, constants and operators constitute an algebraic expression.

Example: 2x+3, 3y+4xy, etc.